

# S3 Demo — Run Steps

Standing the demo up from a clean checkout, in any sandbox

Generated from demo/DEMO\_STEPS.md · repo ams-s3-demo  
Companion docs: apps/README.md (the console and the launch scripts) · repos/README.md (the target repositories, and onboarding one) · demo/DEMO\_TEST\_GUIDE.md (per-scenario rehearsal script)

How to stand this demo up from a clean checkout, in any sandbox. Written for someone who has never run it before — if you already know the repo, you want demo/DEMO\_TEST\_GUIDE.md instead, which is the per-scenario rehearsal script.

## 1. What you are standing up

Five processes, each started by its own script under apps/ , plus the S3 tooling that drives them. You do **not** need all five for every beat.

### TWO FOLDERS, AND THE SPLIT IS THE POINT

repos/ holds the three **target repositories** — PolicyCore, ClaimsPortal, EnrolDirect. Those are what S3 *changes*. apps/ holds the **console and the launch scripts**. Those are what does the changing. Scripts 2–5 below live in apps/ but start code out of repos/ , because starting the demo is a tooling job.

Adding a fourth target is a drop-in: put the source in as repos/<name>/ , add its CRs to the top-level crs/ , add a repos/<name>/.s3targets.json manifest, and keep its regression suite in the top-level tests/ . No edit to any registry file. The manifest contract and the reasoning are in repos/README.md — read that rather than this page for onboarding.

| # | PROCESS                                                                                                                                   | START                      | PORT                       | NEEDED FOR               |
|---|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------|--------------------------|
| 1 | <b>Console</b> — FastAPI + React. The screen you present from. Source in apps/console/ .                                                  | apps/run-console.sh        | 8000 + 5173                | Every beat               |
| 2 | <b>PolicyCore</b> — the client's plan-administration portal (Streamlit), from repos/policycore/ . The window the audience watches change. | apps/run-policycore.sh     | 8501 (open /sl_policycore) | CR-2026-041, CR-2026-042 |
| 3 | <b>Policy-Service</b> — ClaimsPortal contracts side (Python/FastAPI), from repos/claimsportal/policy_service/ .                           | apps/run-policy-service.sh | 8081                       | CR-2026-043              |
| 4 | <b>Claims-Service</b> — ClaimsPortal claims side (Python/FastAPI), from repos/claimsportal/claims_service/ . Start after #3.              | apps/run-claims-service.sh | 8082                       | CR-2026-043              |
| 5 | <b>EnrolDirect</b> — the online enrolment channel (Python/FastAPI), from repos/enroldirect/ . Standalone; needs nothing but the venv.     | apps/run-enroldirect.sh    | 8083                       | CR-2026-045              |

#### OPEN THIS ONE

The console UI is `http://localhost:5173`, not `:8000`. Port 8000 is the API the UI talks to.

See `apps/README.md` for the launch scripts and how each repo maps to a ServiceNow application, and `repos/README.md` for the repositories themselves.

## 2. Prerequisites

| TOOL    | VERSION | NEEDED FOR                                        | CHECK                          |
|---------|---------|---------------------------------------------------|--------------------------------|
| Python  | 3.12+   | Console API, PolicyCore, S3 tooling, ClaimsPortal | <code>python3 --version</code> |
| Node.js | 18+     | Console UI                                        | <code>node --version</code>    |

## 3. First-time setup from base packages

From the repository root. Run once per machine.

```
# 1. Python environment
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# 2. Console UI dependencies
cd apps/console/web && npm install && cd ../../..

# 3. Configuration
cp .env.example .env
# then edit .env - see section 4

# 4. Seed the PolicyCore database
python -m repos.policycore.core.seed

# 5. Confirm the install
python -m pytest -q          # expect: 679 collected, all passing
```

#### SANITY CHECK

If `pytest` passes, the wiring is correct. It exercises the routing registry, the repo-discovery contract, the code-review flow and all four replay targets without needing an LLM or any of the five processes running.

## 4. Pointing it at your LLM

Edit `.env`. Pick *one* provider block.

### Self-hosted / custom model (most sandboxes)

For any endpoint that speaks the OpenAI chat API — an internal gateway, vLLM, LiteLLM, TGI, LM Studio:

```
LLM_PROVIDER=custom
CUSTOM_LLM_BASE_URL=https://llm.internal.example/v1
```

```
CUSTOM_LLM_MODEL=llama-3.3-70b
CUSTOM_LLM_API_KEY=          # optional – blank is fine if your gateway does no auth
```

#### THE BASE URL IS USED EXACTLY AS WRITTEN

Include the version segment your gateway expects, usually `/v1`. Nothing is appended or rewritten, so a wrong URL gives a clean 404 rather than a confusing one. The client requests `<CUSTOM_LLM_BASE_URL>/chat/completions`.

Verify the endpoint is reachable before demo day:

```
python -c "
from common.llm import complete
print(complete('Reply with the single word: ready'))
"
```

## Other providers

| PROVIDER           | LLM_PROVIDER | ALSO SET                           |
|--------------------|--------------|------------------------------------|
| Anthropic          | anthropic    | ANTHROPIC_API_KEY, ANTHROPIC_MODEL |
| OpenAI             | openai       | OPENAI_API_KEY, OPENAI_MODEL       |
| Claude via Bedrock | bedrock      | AWS_REGION, BEDROCK_MODEL          |
| Ollama (local)     | ollama       | OLLAMA_BASE_URL, OLLAMA_MODEL      |

## Replay mode — why the demo survives a bad network

The default is `LLM_MODE=replay`. The code-generation and test-generation beats are served entirely from committed recordings in `s3_enhancement/cache/` and never call your model at all. That is what makes the demo deterministic.

#### SHARP EDGE — READ THIS ONE.

The *narrative* beats (impact analysis, effort estimate, release notes, design doc) go through a different cache that does **not** honour `LLM_MODE`. On a cache miss they call your endpoint live, even in replay mode. Always run `demo/warm_s3_cache.sh` before presenting — otherwise a slow or unreachable model stalls a beat mid-demo.

## 5. Reset before every rehearsal or demo

The demo mutates real source files and the SQLite database. Reset returns everything to the pre-CR baseline. Run these **in this order** — the CR-042 reset must come after `reset_s3.sh`, because that script reseeds the database the CR-042 baseline then builds on.

```
demo/reset_s3.sh          # CR-2026-041 (PolicyCore plan tier) + shared state
demo/reset_s3_endorsement.sh # CR-2026-042 (PolicyCore amendment priority)
demo/reset_s3_claimsportal.sh # CR-2026-043 (ClaimsPortal deductible)
demo/reset_s3_enrolldirect.sh # CR-2026-045 (EnrolDirect prospect access)
demo/warm_s3_cache.sh     # ALWAYS last – reset_s3.sh wipes .cache/llm
```

`reset_s3_endorsement.sh` keeps its pre-reskin filename on purpose: teammates invoke it by name, so only its contents changed with the vocabulary.

Expected final lines:

```
S3 source baseline restored, PolicyCore reseeded, LLM cache cleared, and ticket timeline cleared.
CR-2026-042 source baseline restored, PolicyCore reseeded, and LLM cache cleared.
ClaimsPortal source baseline restored; generated files removed.
EnrolDirect source baseline restored; generated files removed.
```

**THE TWO POLICYCORE RESETS ARE BROKEN RIGHT NOW**

They restore source with `git checkout HEAD -- repos/...`, but the `apps/ → repos/` move is **not committed yet**, so HEAD still carries those files under `apps/` and the checkout fails with *"pathspec did not match"*. **Committing the move fixes it; nothing else does.** The ClaimsPortal and EnrolDirect resets copy from their committed `.baseline/` snapshots and are unaffected. Until then, clear staged proposals by hand with `rm -rf s3_enhancement/out/`.

Or reset from the console

Log in as **Manager** and open `/admin`. Same scopes, no terminal, plus a preview of exactly what each one restores and deletes. It refuses (409) rather than overwriting a dirty tree, there is deliberately no "reset everything" scope, and it reports the breakage above up front as a named `reset_blocked_reason` instead of surfacing a raw git error out of a button.

## 6. Start the applications

One terminal per process. For the PolicyCore CRs you need only the first two.

```
apps/run-console.sh          # terminal 1 - API :8000 + UI :5173
apps/run-policycore.sh      # terminal 2 - portal :8501/sl_policycore

# only for the ClaimsPortal CR (CR-2026-043):
apps/run-policy-service.sh   # terminal 3 - :8081 (start before claims)
apps/run-claims-service.sh   # terminal 4 - :8082

# only for the EnrolDirect CR (CR-2026-045):
apps/run-enroldirect.sh      # terminal 5 - :8083
```

A manager can also see and control processes 2–5 from `/admin` without a terminal. Not the console itself — it cannot restart the process serving the request, and naming it is a 400 rather than an attempt.

Then open `http://localhost:5173` and log in. Any roster name works; passcodes are in `common/roster.py`.

| LOGIN      | PASSCODE | ROLE                                 |
|------------|----------|--------------------------------------|
| Ravi Kumar | 1001     | Engineer — App Support, PolicyCore   |
| Priya Nair | 1003     | Engineer — App Support, ClaimsPortal |
| Manager    | 9000     | Manager view                         |

## 7. Optional — seed the routing beat

With the console API already running, this creates a ticket that carries a ServiceNow Configuration Item, so the console can route it to an owning team before any AI step runs.

```
# Default: routes to BillingGateway – an application with an owning team and
# NO repo here. Routing succeeds; automation correctly stays off.
demo/seed_problem_record_ticket.sh

# The other half: routes to a team AND offers the CR to run against it.
SEED_CI=ClaimsPortal SEED_BUSINESS_SERVICE="Claims Management" \
demo/seed_problem_record_ticket.sh
```

Re-running with a different `SEED_CI` re-routes the same ticket; you do not need to reset between the two.

## 8. The demo flow

| #  | BEAT                                                                                | WHAT TO SAY IT PROVES                                                                                                                                                                                                           |
|----|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0  | <b>The board already has a ticket for a CR nobody seeded</b> — AMS-1045, unassigned | A <code>.md</code> file dropped into <code>crs/</code> opens a ticket by itself, keyed off the CR id, and lands unassigned so it routes to the manager. That is the onboarding story being literally true rather than described |
| 1  | Open the ticket; routing panel appears above the analysis                           | The CI resolved to an application, owning team and repo by table lookup — no model call, nothing to confirm                                                                                                                     |
| 2  | Impact analysis + effort estimate                                                   | Vague tickets get a clarifying question first, rather than a confident guess                                                                                                                                                    |
| 3  | Generate code                                                                       | Only the files the relevance funnel selected are sent — the token panel shows scoped vs naive cost                                                                                                                              |
| 4  | <b>Review file by file:</b> Ask, Apply this file, Reject                            | Developers accept or reject one file at a time; a rejection records a reason to the ticket's audit trail and is excluded from Apply                                                                                             |
| 5  | Apply, then look at PolicyCore on <code>:8501/sl_policycore</code>                  | The client's running application changed                                                                                                                                                                                        |
| 5b | <b>Source control panel:</b> branch → commit → push                                 | The change lands on a feature branch cut off <code>main</code> before anything is written, the commit is gated on the tests passing, and the push hands off to the pipeline — the flow, not a direct edit to <code>main</code>  |
| 6  | <b>Revert</b> (per file or all)                                                     | Anything applied can be undone without a full demo reset                                                                                                                                                                        |
| 6b | <b>Design doc: change map + Download PDF</b>                                        | The hand-off document carries a diagram of what the change touches, and leaves as a real PDF you can attach to the ticket                                                                                                       |
| 7  | <b>Draft test scenarios</b> , edit one, approve the plan                            | QA reviews <i>what will be checked</i> , in prose traced to the CR's acceptance criteria, before any test code exists — and can change it                                                                                       |
| 8  | Generate tests, run them, then the seeded-bug check                                 | The generated tests actually catch a deliberately introduced bug                                                                                                                                                                |
| 9  | <b>Run the regression suite</b>                                                     | A human-authored suite the AI cannot write to still passes — the CR cost nothing that already worked                                                                                                                            |

| #  | BEAT                                                                                                 | WHAT TO SAY IT PROVES                                                                                                                                                                           |
|----|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | <b>Build the traceability matrix</b>                                                                 | Every acceptance criterion, the scenarios planned for it, the tests that ran, and the result — the artifact an auditor asks for                                                                 |
| 11 | Design-doc drift check                                                                               | Documentation drift is detected automatically after Apply, not by remembering to press a button                                                                                                 |
| 12 | <b>Release notes — three audiences + the derived deployment &amp; rollback plan</b>                  | One note per reader (client / ops / user guide), and a deploy order computed from the change's own service graph                                                                                |
| 13 | <b>Download the release record, attach it to the ticket</b>                                          | Everything the run proved, in one PDF — including what it could <i>not</i> prove                                                                                                                |
| 14 | <b>Optional close, as Manager:</b> <code>/admin</code> — service state, reset scopes, onboard a repo | Everything that used to need a terminal mid-demo, behind a server-side manager check. Good place to answer "what would it take to point this at our repo?" with a screen instead of a paragraph |

#### ONE UI CHANGE TO EXPECT IF YOU REHEARSED ON AN OLDER BUILD

Artifacts now open in a **modal** rather than rendering inline — release notes, deployment plan, design doc, scenario table, test checklists, mutation diff, traceability matrix. Action buttons, verdict lines and token-cost lines stay in the main flow, and the left-hand stage rail is unchanged. This came straight from the client's "there is a lot of data on the screen... open it in a pop-up" feedback, so it is worth naming as *their* change when it comes up.

On beats 12-13: the deployment order is **derived**, not drafted — on CR-2026-043 the plan puts `policy_service` before `claims_service` because `claims_service` calls it, and says why. The release record is assembled from what the run actually produced; its "Not evidenced by this release" block is the part worth pausing on, because a release document that only lists successes is marketing. **Attach to ticket** is honest about the demo default: with `JIRA_MODE=replay` there is no Jira to attach to, so the beat records the intent on the ticket timeline and says the upload was simulated. Set `JIRA_MODE=live` and it uploads for real.

On beat 5b: say plainly that the git flow is **modelled, not executed** — the panel says so on screen and the release record repeats it under "Not evidenced by this release". Nothing runs git and no remote is contacted. That is deliberate: the target apps live inside this repo and the reset scripts restore the baseline from `HEAD`, so a real commit would make them start restoring the CR instead. The point of the beat is the *shape* of the flow — branch before edit, commit gated on green tests, pipeline on push — which is what a reviewer asks about when they see an AI editing code.

Two things are worth clicking rather than describing. Press **Commit to branch** before running the tests: it refuses, and names the reason, because the gate is computed server-side from the ticket's own test results — the console cannot assert "tests passed". And open **What a real integration would have run** for the git transcript, which grows one step at a time as you take each step. If you Revert everything afterwards, the branch shows as *abandoned* rather than disappearing, and a commit already made is not unmade — in a real repo the honest undo at that point is a revert commit, not a rewritten history.

On beat 6b: the change map is **derived, not drawn by the model** — services, layers and the cross-service arrow are read from the changed-file set, so it costs no LLM call and needs no cache warming. The **NEW** badge comes from git (the file is absent from `HEAD`), which is why `claim_rules.py` carries one on CR-2026-043 and nothing does on CR-2026-042. The PDF is rendered server-side by headless Chromium; if

`playwright install chromium` has not been run on the demo machine the endpoint answers 503 and the console silently falls back to the browser's own print-to-PDF, so the button always does something.

Beats 7, 9 and 10 are the QA-facing half of the tests stage. Two things worth saying out loud when showing them:

- The regression suites — `tests/test_regression_policycore.py` (20 tests), `tests/test_regression_policycore_enrolment.py` (19), `tests/test_regression_claimsportal.py` (5) and `tests/test_regression_enroldirect.py` (24) — appear in **no** target's `testgen_allowlist` or `codegen_allowlist`, and `tests/test_s3_testrun.py` asserts it. The pipeline physically cannot write to them, which is what makes "the pre-existing tests still pass" a result rather than a claim. They also live in the top-level `tests/` rather than under a repo root, because anything ending `.py` under `repos/<name>/` joins the codegen candidate pool.
- In the matrix, only the scenario→test column is inferred, and it is deliberately conservative: an ambiguous pairing renders as "no automated test" rather than guessing. On CR-2026-043 two criteria legitimately land there — that is the honest answer, and a good moment to make the point that the tool reports gaps instead of hiding them.

For the full per-scenario talk track and the fallback ladder, see `demo/DEMO_TEST_GUIDE.md`.

## 9. Troubleshooting

| SYMPTOM                                                                    | CAUSE AND FIX                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>FOREIGN KEY</code> constraint failed during a reset or seed          | An old baseline whose <code>wipe_db()</code> predates the amendments table. Fixed in the current scripts. If you hit it on an older checkout: <code>rm -f data/mockapp.db</code> , then re-run the reset.                                                                        |
| <code>pathspec 'repos/policycore/app.py' did not match</code> from a reset | Known: the <code>apps/ → repos/</code> move is uncommitted, so HEAD does not have those paths. Commit the move. Nothing else fixes it; see section 5.                                                                                                                            |
| <code>codegen</code> returned unexpected file set                          | A repository under <code>repos/</code> was renamed or moved. The replay recordings contain the exact paths, both as file keys and inside the generated code's own <code>import</code> statements. Restore the directory name — see the warning in <code>repos/README.md</code> . |
| A repo you dropped into <code>repos/</code> never shows up                 | Either it has no <code>.s3targets.json</code> , or the console has not restarted since one was written. A manifest that exists but cannot be parsed <i>raises at import</i> rather than being skipped — so silence means "no manifest", not "bad manifest".                      |
| A beat hangs, or fails mentioning your LLM URL                             | A narrative beat missed its cache and called your model. Run <code>demo/warm_s3_cache.sh</code> . Confirm the endpoint with the one-liner in section 4.                                                                                                                          |
| Applied change crashed the portal                                          | Expected and handled — the console shows the migration traceback with a one-click fix. You can also press <b>Revert all</b> .                                                                                                                                                    |
| Console UI loads but every call 401s                                       | Not logged in, or the API on :8000 is not running. Check terminal 1.                                                                                                                                                                                                             |
| Claims-Service errors on a claim                                           | Policy-Service on :8081 is not up. Claims calls policy over HTTP; start #3 first.                                                                                                                                                                                                |

| SYMPTOM             | CAUSE AND FIX                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port already in use | <code>lsof -ti:8000   xargs kill</code> (repeat per port). Note that <code>--reload</code> is deliberately not used on the API: the reloader restarts mid-beat when codegen writes to the tree. |

## 10. Pre-demo checklist

- ☐ `python -m pytest -q` passes
- ☐ `.env` has your provider block filled in, and the section-4 one-liner returned a reply
- ☐ All four reset scripts ran clean, in order — or you know which of them is currently blocked and why
- ☐ `demo/warm_s3_cache.sh` ran *after* the resets
- ☐ Console reachable at `:5173`; you are logged in
- ☐ PolicyCore reachable at `:8501/sl_policycore` in a second window
- ☐ For the ClaimsPortal CR: `:8081` and `:8082` both responding
- ☐ For the EnrolDirect CR: `:8083` responding, and a prospect still refused at `POST /api/eligibility/check` — that refusal *is* the baseline
- ☐ You have walked beats 0–14 once, end to end, on this machine

The three target repositories live under `repos/` — what S3 changes; the console and the launch scripts live under `apps/` — what does the changing. The AI pipeline is in `s3_enhancement/`, shared clients in `common/`, change requests in `crs/`, and presenter scripts in `demo/`. Do not rename a directory under `repos/`: a repo's path is a scoring input and it is baked into the committed recordings, so moving one is a two-part rewrite, not a `mv`. See `repos/README.md`.